

8 Humans have introduced new species to the island of St Helena.

Some of these species have become invasive and have reduced the biodiversity on this island.

These invasive species include cats, rabbits and rats.

- (a) The St Helena plover is a bird endemic to this island. It lays its eggs in a nest on the ground.

The photographs show a nest and offspring of a St Helena plover.



(Source: © Snapper Nick / Alamy Stock Photo)

Nest of a St Helena plover



(Source: © Buiten-Beeld / Alamy Stock Photo)

St Helena plover chick

- (i) Suggest why the population of the St Helena plover decreased after the introduction of cats and rats to the island.

(2)

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- (ii) Conservation programmes resulted in the population of the St Helena plover increasing from 200 adult birds in 2006 to 560 adults in 2016.

Calculate the mean rate of increase in this population.

(1)

..... number of birds per year

(b) The St Helena gumwood tree is native to the island of Saint Helena.

The photograph shows part of a St Helena gumwood tree.



(Source: © Prisma by Dukas Presseagentur GmbH / Alamy Stock Photo)

The St Helena gumwood tree is critically endangered.

Rabbits have contributed to this situation by eating the shoots of young trees.

A breeding programme increased the population of the gumwood trees from two trees in 2009 to 6000 in 2019.

- (i) Describe how this St Helena breeding programme could have been carried out.

(2)



(ii) Explain **one** way that young trees could be protected from rabbits.

(2)

(iii) Scientists have proposed storing seeds from St Helena gumwood trees in seed banks to aid the conservation of this endangered species.

Describe how a seed bank would successfully select, prepare **and** store seeds from these trees to aid the conservation of this endangered species.

(4)

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(c) A study compared the biodiversity of two habitats on the island of St Helena.

The table shows the data obtained from habitat 1.

Species	Number of individuals (n)	$n(n-1)$
A	747	557 262
B	8	56
C	75	5 550
D	12	132
E	63	
	$N = 905$	$\sum n(n-1) =$



- (i) An index of diversity (D) is calculated using the formula:

$$D = \frac{N(N-1)}{\sum n(n-1)}$$

Calculate the index of diversity for habitat 1.

Use the table and formula to help you.

Give your answer to **two** decimal places.

(3)

Answer

- (ii) Habitat 2 has an index of diversity (D) of 2.80

A student concluded that the second habitat had a higher biodiversity.

State whether you agree or disagree with this conclusion. Give a reason for your answer.

(1)

(Total for Question 8 = 15 marks)

TOTAL FOR PAPER = 80 MARKS

